



Investigation on the Acoustic Performance of Multiple Helmholtz Resonator Configurations

K. Mahesh¹ · R. S. Mini¹

Received: 31 October 2020 / Accepted: 23 March 2021 / Published online: 13 April 2021
© Australian Acoustical Society 2021

Abstract

Helmholtz resonator is considered and widely accepted as a basic acoustic model in engineering applications and research. In this paper, the normal incidence sound absorption characteristics of series and parallel configurations of Helmholtz resonators is studied analytically, numerically and experimentally. The proposed analytical model for series configuration of HRs comprises of Johnson–Champoux–Allard model and transfer matrix method while parallel configuration of HRs is described using parallel transfer matrix method. The results from proposed analytical models fit well with the finite element method (FEM) results obtained from COMSOL multiphysics. Incorporation of parallel configuration and proper tuning of geometric parameters helps to overcome the trade-off between broad band sound absorption and minimum space utilization. Also, the experimental observations of one of the parallel configuration substantiates the FEM results. Moreover, the FEM models are more accountable for the variation in neck position and also provide better visualization of acoustic absorption with frequency.

Keywords Helmholtz resonator · Sound absorption · Parallel transfer matrix method · Johnson–Champoux Allard model · Finite element method

1 Introduction

Helmholtz resonators (HR) are actively used as the noise reduction device in silencers, ducted systems like heating, ventilation, air conditioning (HVAC), tunnels, etc. [1]. The normal incidence sound absorption coefficient is a major parameter used to define the absorption characteristics of HR. The classical lumped parameter approach is the usual method to describe it. But it has frequency limitations, and it loses accuracy for resonator length $L > \lambda/16$ [2–6]. A number of researchers also used one-dimensional analytical approaches to study the absorption behavior of HRs with better approximations [6–8].

Geometric parameters of HR, neck and cavity, decide its absorption behavior. However, a single HR is not suitable to mitigate multi-tonal noises due to its narrow band sound absorption characteristics. Several studies dealt with ways for

enhancing sound absorption and transmission loss properties of Helmholtz resonators [9–13]. Selamet et al. [14] observed variation in resonance frequency and transmission loss with variation in length, shape and perforations on the extended neck of HR. A significant improvement in sound absorption was noticed with tapering of neck away from the cavity [15]. The addition of porous material in HR cavity was another method used to improve the absorption properties of HR [16]. Conversion of Helmholtz resonator neck into spiral form also helped to achieve high sound absorption at low frequency with less space consumption [17,18].

Rather than individual HR, arrangement of HRs in combinations yielded improved sound absorption characteristics [11,19,20]. Sang et al. [11] observed that parallel arrangement of HRs broadened the bandwidth of absorption along with improvement in resonance peak, whereas series arrangement of resonators enhanced the absorption coefficient at resonant frequency. The sound absorption of Helmholtz resonator array in the low frequency regime was theoretically predicted by Kim et al. [21]. A dual HR with two necks and two cavities produced two low frequency resonance peaks [19], while a periodic array of dual HRs produced multiple absorption peaks corresponding to its resonance frequencies [20]. Tang et al. [22] enhanced the low

✉ R. S. Mini
miniranjithrs@gmail.com

K. Mahesh
maheshnilambur2012@gmail.com

¹ Department of Mechanical Engineering, College of Engineering Trivandrum, Thiruvananthapuram, Kerala 695016, India

frequency sound absorption by a coupled resonator with a front and a rear resonator. In the meantime, HR was found integrated with microperforated panel (MPP) to attenuate broad band of sound [23–25].

Recently, due to its geometric simplicity, several researchers used HR as the fundamental building block of acoustic metamaterials [26–37]. Acoustic metamaterials are found to exhibit excellent sound absorption and sound transmission behaviors [27,28,30,31,34–39]. It is well known that HRs are particularly useful in the low frequency range irrespective of its narrow band sound absorption. By incorporating them as the basic units of acoustic metamaterials, broad band sound absorption was attained in the low frequency regime [30,32,33]. In this scenario, a thorough analysis of various influence factors on the resonant frequency was done by Li et al. [40] using finite element method. Later, Cai et al. [41] proposed noise attenuation capacity as an additional parameter to assess the acoustic performance of HR. Thus, it is understood that HRs are still in the forefront of acoustics research, which requires further exploration.

Though, Helmholtz resonator is a hot topic in acoustic research, the lacuna of detailed investigations of different Helmholtz resonator configurations analytically using transfer matrix techniques and numerically using finite element method is addressed in detail in the present research. Here, simple and effective semi-phenomenological models proposed by Johnson et al. [42] and Champoux and Allard [43] are used to model the viscous and thermal dissipation in HR. Using JCA method and transfer matrix method (TMM) [44,45], the absorption characteristics of series combination of HRs are studied whereas for parallel combination, parallel transfer matrix method (P-TMM) is adopted. The P-TMM is proved to be adequate for modeling the parallel combination of resonators with acceptable accuracy [46,47]. To the authors knowledge, it is the first time the acoustic performance of parallel configuration of resonators is thoroughly emphasized using P-TMM and JCA methods. The P-TMM is such a flexible method to model parallel combination of HRs and JCA which adds more fidelity to the results as it is accurate at low and high frequencies when the wavelength of sound is much greater than the characteristic dimension of the resonator [48]. Also a detailed comparison and error analysis between the numerical method and analytical method is enclosed in this work. The paper is structured as follows: Sect. 2 describes about the analytical and numerical methodology adopted for the determination of absorption coefficient. The absorption characteristics of series and parallel configurations of HRs, detailed error analysis between FEM and analytical results, effect of offset neck position on absorption characteristics and experimental realization of a HR absorber are described in Sect. 3. Finally, the concluding remarks are dealt in Sect. 4.

2 Theoretical Modeling

The present research investigates the absorption characteristics of conventional Helmholtz resonator and its series and parallel configurations using analytical, numerical and experimental methodologies. The corresponding theoretical aspects are described in subsequent sections.

2.1 Analytical Modeling

The analytical modeling of series and parallel configurations of HRs is described in this section. The Johnson–Champoux–Allard (JCA) model is used to consider the losses in the resonator's neck and the acoustic behavior of neck and cavity are represented by 2×2 transfer matrices. The absorption characteristics of series configuration of HRs are predicted by classical transfer matrix method, while parallel configurations of HRs are modeled by P-TMM.

2.1.1 HR Model

Conventional Helmholtz resonator with a cylindrical neck backed by an air cavity is considered in this study. The schematic view of the Helmholtz resonator is shown in Fig. 1, where d_n is the neck diameter, l_n is the neck length, l_c is the cavity length, and d_c is the cavity diameter. The JCA model with TMM is adopted to study the absorption characteristics of HRs. TMM is considered as a powerful method to predict the acoustical behavior of multilayered structures which are stacked in series [45,46]. It relates the sound pressure and particle velocity of the upstream and downstream of each layer of HR (neck and cavity). Transfer matrix of a Helmholtz resonator (T_{HR}) can be written as the product of transfer matrices of neck (T_n) and cavity (T_c), ie.

$$[T_{HR}] = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} = [T_n][T_c], \quad (1)$$

where,

$$[T_c] = \begin{bmatrix} \cos(k_c l_c) & i Z_c \sin(k_c l_c) \\ \frac{i}{Z_c} \sin(k_c l_c) & \cos(k_c l_c) \end{bmatrix} \quad (2)$$

and

$$[T_n] = \begin{bmatrix} \cos(k_n l_n) & i Z_n \sin(k_n l_n) \\ \frac{i}{Z_n} \sin(k_n l_n) & \cos(k_n l_n) \end{bmatrix}. \quad (3)$$

For HR cavity, the characteristic impedance $Z_c = \rho_0 c_0$ and the wave number $k_c = \omega / c_0$, where ω is the angular frequency, c_0 is the velocity of sound in air and ρ_0 is the density of air. The effective wave number (k_n) and characteristic impedance (Z_n) for the neck is written as [45],

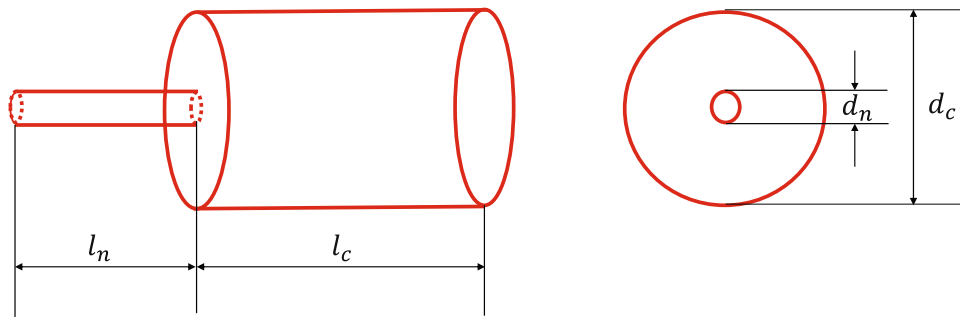


Fig. 1 Schematic diagram of the Helmholtz resonator

$$k_n = \omega \sqrt{\rho_{eq} / K_{eq}}, \quad (4)$$

$$Z_n = \sqrt{\rho_{eq} K_{eq}}, \quad (5)$$

where ρ_{eq} is the complex effective fluid density and K_{eq} is the complex effective bulk modulus. These two acoustic properties describe the viscothermal dissipative effects of the neck, which are obtained from the JCA model. The model is based on five intrinsic properties, i.e., flow resistivity σ , porosity ϕ , tortuosity α_∞ , viscous characteristic length Λ and thermal characteristic length Λ' . These properties chosen for HR are given in Table 1. Among these, the tortuosity depends on the correction length of the neck. The correction length accounts for the mass loading associated to the sound radiation of the neck and distortion of acoustic flow around the resonator surface. The correction length of the neck (ε) is,

$$\varepsilon = 0.48\sqrt{s_n}(1 - 1.14\sqrt{s}), \quad (6)$$

where $s_n = \pi r_n^2$ is the cross sectional area of neck and $s = (\pi r_n^2)/(\pi r_c^2)$ is the open area ratio. Here r_n and r_c are the radius of neck and cavity, respectively. The above relation for correction length is applicable for $s < 0.16$ only [45].

The complex density (ρ_{eq}) and bulk modulus (K_{eq}) of the equivalent fluid in the neck are determined using the relation,

$$\rho_{eq} = \frac{\rho_0}{\phi} \left(\alpha_\infty + \frac{\sigma \phi}{j\omega\rho_0} F(\omega) \right) \quad (7)$$

and

$$K_{eq} = \frac{1}{\phi} \frac{\gamma P_0}{\gamma - (\gamma - 1) \left(1 + \frac{8\eta}{j\omega\rho_0 N_{Pr} \Lambda'^2} G(\omega) \right)^{-1}}, \quad (8)$$

where γ is the ratio of specific heats, P_0 is the atmospheric pressure, η is the dynamic viscosity of air, and N_{Pr} is the Prandtl number. $F(\omega)$ and $G(\omega)$ are the correction functions for complex effective fluid density and complex effective bulk modulus, respectively, which are expressed as [45],

$$F(\omega) = \sqrt{1 + \frac{j\omega 4\eta\rho_0\alpha_\infty^2}{\phi^2\sigma^2\Lambda^2}}, \quad (9)$$

and

$$G(\omega) = \sqrt{1 + \frac{j\omega\rho_0 N_{Pr} \Lambda'^2}{16\eta}}. \quad (10)$$

2.1.2 Series Configuration of HRs

In series configuration, HRs are arranged successively (Fig. 2a). Using the individual transfer matrix of neck and cavity, the overall transfer matrix of series configuration of 'n' number of HRs is formulated as,

$$T^s = \prod_{i=1}^n T_n^i T_c^i. \quad (11)$$

From the overall transfer matrix, the reflection coefficient, R , is calculated as per the relation,

$$R = \frac{T_{11}^s - T_{21}^s Z_0}{T_{11}^s + T_{21}^s Z_0}, \quad (12)$$

Table 1 Parameters of JCA model, where R_h is the hydraulic radius and η is the dynamic viscosity of the fluid medium [45]

Cross shape	sectional	Viscous characteristics Λ	charac-length	Thermal characteristics length Λ'	Tortuosity α_∞	Static flow resistivity σ
Circle		R_h		R_h	$1 + (2\varepsilon/l_n)$	$8\eta/(\phi R_h^2)$

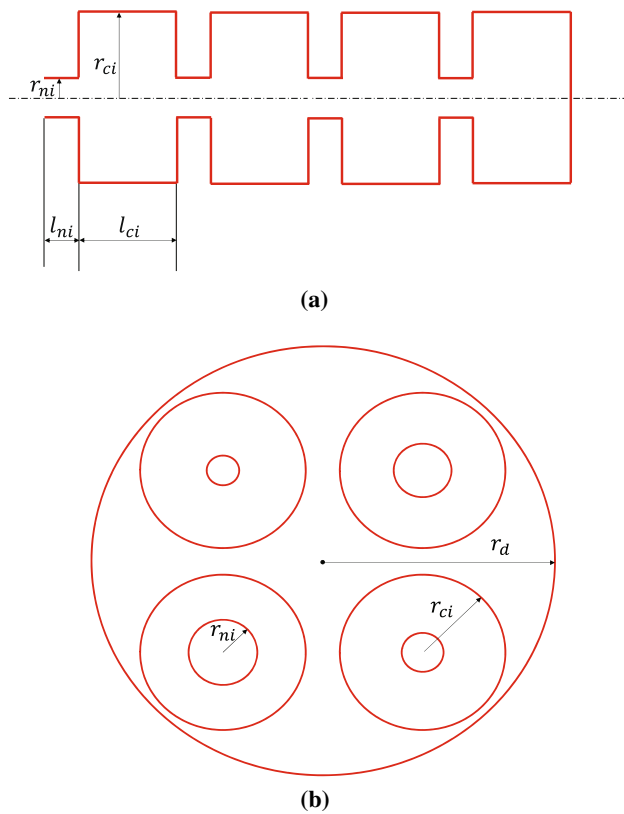


Fig. 2 Schematic diagram of **a** series and **b** parallel configuration of HRs

and from which the normal incidence sound absorption coefficient α is determined as,

$$\alpha = 1 - |R|^2. \quad (13)$$

2.1.3 Parallel Configuration of HRs

In this study, parallel assembly of four HRs is considered. Its schematic view is shown in Fig. 2b, where r_{ni} is the neck radius, r_{ci} is the cavity radius, and r_d is the panel radius. P-TMM, an efficient tool used to assemble the transfer matrices of elements which are arranged in parallel direction, is used for modeling. According to Verdiere et al. [46], certain conditions need to be satisfied for the application of P-TMM. (1) Only plane waves propagate upstream and downstream of the periodic construction; (2) only normal incidence plane waves propagate in the construction; (3) no pressure diffusion exists between adjacent parallel elements, (4) the wavelength is much larger than the HR assembly, and (5) each element can be represented by a 2×2 transfer matrix. In the present study, the conditions (2)–(5) are satisfied since: (i) size of the HR assembly is very much lower than the lowest wavelength of interest, (ii) the HR neck is modeled using the equivalent fluid model (JCA model), (iii) the HR is modeled using 2×2

transfer matrices. However, the condition (1) is physically unrealistic around the HR resonance frequency in the vicinity of the HR neck. From the transfer matrix of HR (T_{HR}), the corresponding admittance matrix (Y_{HR}) is written as [46],

$$Y_{HR} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \frac{1}{t_{12}} \begin{bmatrix} t_{22} & t_{21}t_{12} - t_{22}t_{11} \\ 1 & -t_{11} \end{bmatrix}. \quad (14)$$

Then the transfer matrix (T^P) corresponding to the parallel construction of HRs is written as [46],

$$T^P = \frac{-1}{\sum r_i Y_{21}^i} \begin{bmatrix} \sum r_i Y_{22}^i & -1 \\ \sum r_i Y_{11}^i - \sum r_i Y_{12}^i \sum r_i Y_{21}^i & -\sum r_i Y_{11}^i \end{bmatrix}, \quad (15)$$

where Y^i are the elements of admittance matrix of i^{th} resonator and surface area ratio of i^{th} HR is given by, $r_i = (\pi r_{ci}^2)/(\pi r_d^2)$. Then from the overall transfer matrix the reflection coefficient R and absorption coefficient α are determined as follows,

$$R = \frac{T_{11}^P - T_{21}^P Z_0}{T_{11}^P + T_{21}^P Z_0}, \quad (16)$$

and

$$\alpha = 1 - |R|^2. \quad (17)$$

2.2 Numerical Modeling

In this work, the three-dimensional simulations are carried out using COMSOL multiphysics software. For absorption studies, numerical model of the impedance tube setup (as the wave guide) with absorber is considered. The propagation of sound through the impedance tube is governed by the acoustic wave equation. In frequency domain, it reduces to a Helmholtz equation which is of the form, [49]

$$\Delta p + k^2 p = 0, \quad (18)$$

where k is the wave number and p is the acoustic pressure. Wave number $k = \omega/c_0$, where ω is the angular frequency and c_0 is the velocity of sound in the medium. Walls of the impedance tube and its right end are considered sound hard [50], which yields total reflection (due to infinite impedance), for which

$$\frac{\partial p}{\partial \mathbf{n}} = 0. \quad (19)$$

Plane harmonic wave excitation is given at the left end of the waveguide, where radiation boundary condition is incorporated. The domain inside the impedance tube is selected

as air with density $\rho_0 = 1.16 \text{ kg/m}^3$ and velocity of sound $c_0 = 346 \text{ m/s}$. The necks of HRs are modeled as equivalent fluid with complex density and bulk modulus values determined from the JCA model. Under JCA descriptions, the wave number term in Eq. (18) is replaced by the effective wave number given in Eq. (4). The JCA parameters used for the modeling are given in Table 1. The tortuosity and the porosity being implicit in the problem geometry itself; hence, they are set as 1. The entire domain is discretized using tetrahedral elements. The highest frequency used for the study is 2000 Hz, and the corresponding wavelength $\lambda_{\min} = c/f_{\max} = 346/2000 = 173 \text{ mm}$. The minimum element size chosen is 0.1 mm which is about $\lambda_{\min}/1730$, and the maximum element size chosen is 10 mm which is about $\lambda_{\min}/17$. Adequate meshing is achieved by choosing such a minimum element size in the neck region. The acoustic response is numerically simulated in the frequency range of 2–2000 Hz with a frequency step of 2 Hz. In all numerical models depicted in this paper, the aforementioned meshing and frequency ranges are followed. The pressure response at two points, which corresponds to the microphone positions is extracted for analysis (Ref. Fig. 3). The microphones are separated by a distance s . The distance between first microphone and sample is x_1 and that between second microphone and sample is x_2 . As per ISO 10534-2 standard [51], the sound absorption coefficient under normal incidence is calculated using transfer function (H_{12}),

$$H_{12} = \frac{p_2}{p_1} = \frac{e^{ikx_2} + Re^{-ikx_2}}{e^{ikx_1} + Re^{-ikx_1}}, \quad (20)$$

where p_1 and p_2 are the pressure responses at microphone positions 1 and 2, respectively. The reflection coefficient (R) and absorption coefficient (α) are calculated using the following relations,

$$R = \frac{H_{12} - e^{-iks}}{e^{iks} - H_{12}} e^{2ikx_1} \quad (21)$$

and

$$\alpha = 1 - |R|^2. \quad (22)$$

3 Results and Discussion

The absorption characteristics of various series and parallel configurations of HRs is studied analytically and numerically. In both series and parallel configurations, array of similar as well as dissimilar HRs are considered. The FEM simulation results and analytical results are compared and contrasted. Experimental validation of one of the results is also described.

3.1 Absorption Characteristics of an HR

The absorption characteristics of conventional HR is considered as the preliminary study. An HR with neck diameter $d_n = 3 \text{ mm}$, neck length $l_n = 8 \text{ mm}$, cavity length $l_c = 12 \text{ mm}$ and cavity diameter $d_c = 50 \text{ mm}$ is chosen for the study. Analytically predicted absorption coefficients and the FEM simulation results of this particular HR is compared in Fig. 4. It is observed that the analytical results are in good agreement with the FEM simulation results. It is also noted that the particular HR shows around 99% absorption at resonance frequency of 286 Hz. Also from the corresponding pressure plot (Fig. 5), one can visualize that the sound pressure dissipation is maximum at the resonance condition.

3.2 Series Configuration of HRs

In series configuration, HRs are arranged successively one after the other. The arrangement is similar to the dual HR model (neck-cavity-neck-cavity model) proposed by Xu et al. [19]. Here instead of two necks and two cavities, four necks and four cavities are arranged alternatively (Ref. Fig. 2a). Hence a maximum of four resonance peaks are

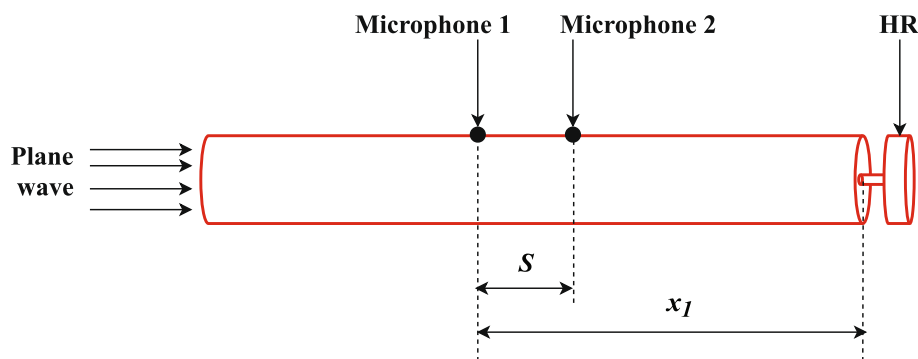


Fig. 3 Schematic diagram of the numerical model (impedance tube setup with HR as test sample)

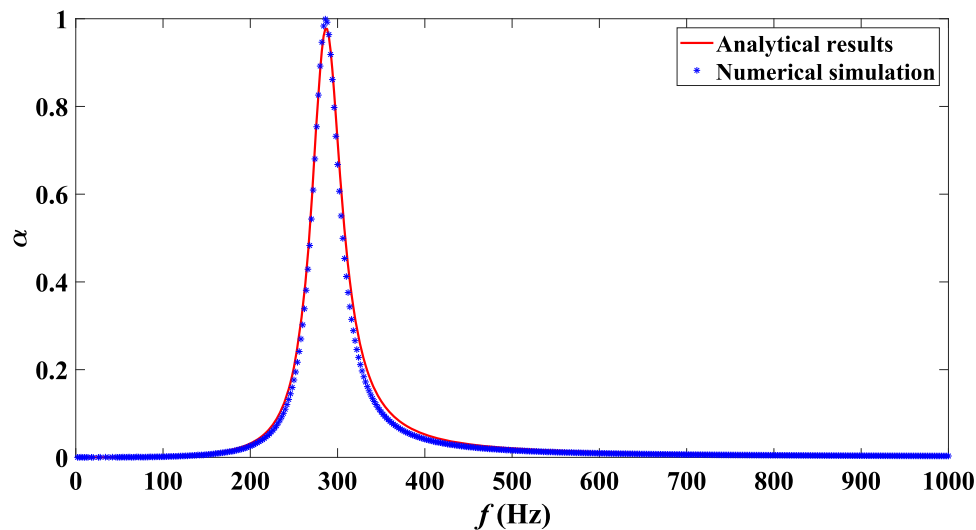


Fig. 4 Comparison of analytical predictions and the FEM simulation results of HR

freq(143)=286 Hz Volume: Absolute pressure (Pa)

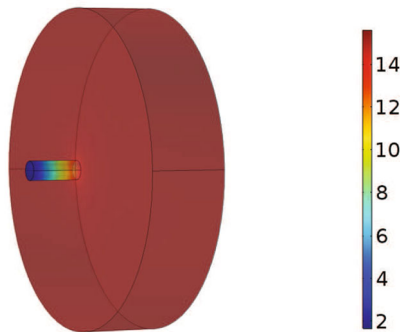


Fig. 5 The pressure plot of HR at its resonance frequency

possible with this configurations, irrespective of the type of resonators used. To verify this concept, absorption characteristics of series configuration of four similar as well as dissimilar HRs are separately analyzed. The corresponding isometric views of the models are depicted in Fig. 6.

3.2.1 Series Configuration of Similar HRs

In series configurations of similar HRs, four HRs of identical dimensions are chosen. The dimensions of HRs are: neck diameter $d_n = 2$ mm, neck length $l_n = 8$ mm, cavity length $l_c = 12$ mm and cavity diameter $d_c = 25$ mm. Absorption coefficient obtained for series arrangement of similar HRs from the analytical model and FEM simulations is compared in Fig. 7. It is observed that the analytical results agree well with the FEM simulation results. Here the four HRs coupled together as a single structure and exhibits better absorption corresponding to the four resonance frequencies. The coupling mechanism is well defined in Ref.[19].

3.2.2 Series Configuration of Dissimilar HRs

Next, absorption characteristics of series configuration of dissimilar HRs is studied. Here two different cases are studied. In the first case, the HRs are arranged in the increasing order of neck radius. The neck radius chosen are 1 mm, 1.5 mm,

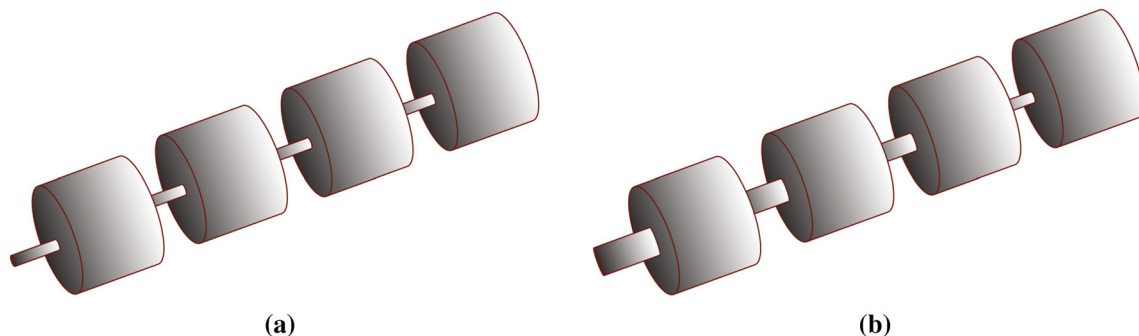


Fig. 6 The isometric view of series configuration of **a** similar HRs and **b** dissimilar HRs

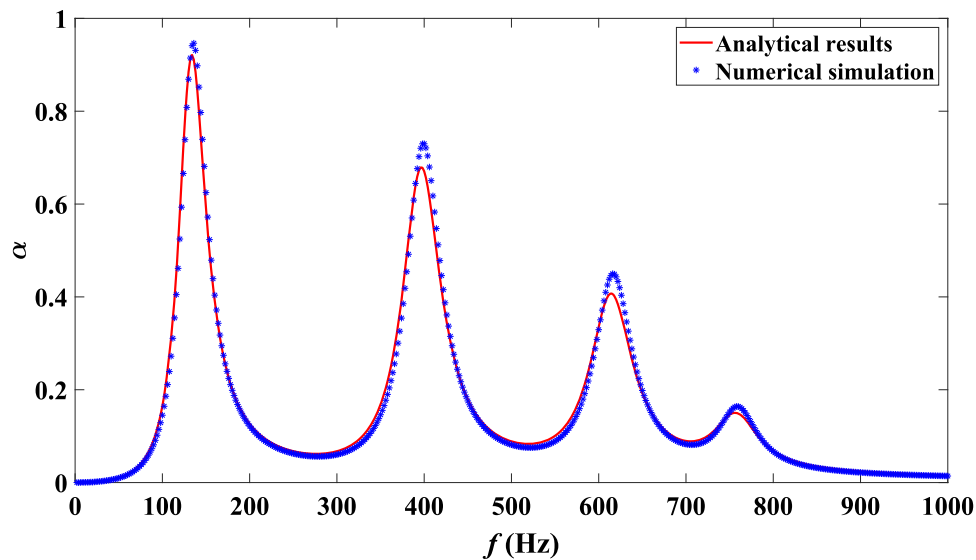


Fig. 7 Absorption coefficient obtained from analytical model and FEM simulations for the series arrangement of similar HRs

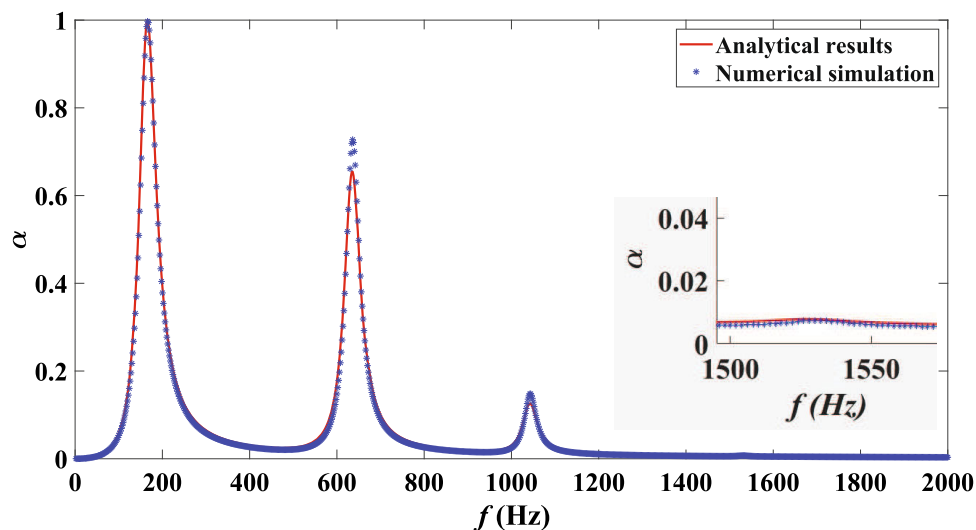


Fig. 8 Absorption coefficients obtained from analytical model and FEM simulation for series arrangement of dissimilar HRs in the ascending order of neck diameter. The magnified view of the fourth resonance peak is shown as inset

2 mm and 2.5 mm. The other dimensions chosen are same as that of the previous case. Figure 8 compares the results obtained from analytical method and FEM simulation for this model, which shows well agreement. This configuration also has four absorption peaks corresponding to the resonance frequencies. But the fourth peak shows very small absorption coefficient of 0.0078 at 1528 Hz. In the second case, HRs are arranged in the decreasing order of neck radius. Figure 9 shows the absorption characteristics of this configuration, which shows more than 90% absorption corresponding to the resonant frequencies. Also this configuration exhibits resonance at comparatively lower frequencies than the one with ascending arrangement of necks.

3.3 Parallel Configuration of HRs

Successive arrangement of HRs increase the overall thickness of the absorber. By adopting parallel arrangement of HRs, the thickness of the absorber can be reduced considerably. In this section, the absorption characteristics of parallel configuration of similar as well as dissimilar HRs are considered.

3.3.1 Parallel Configuration of Similar HRs

In the first case, parallel assembly of similar HRs is modeled. The dimensions of all HRs are: neck diameter $d_n =$

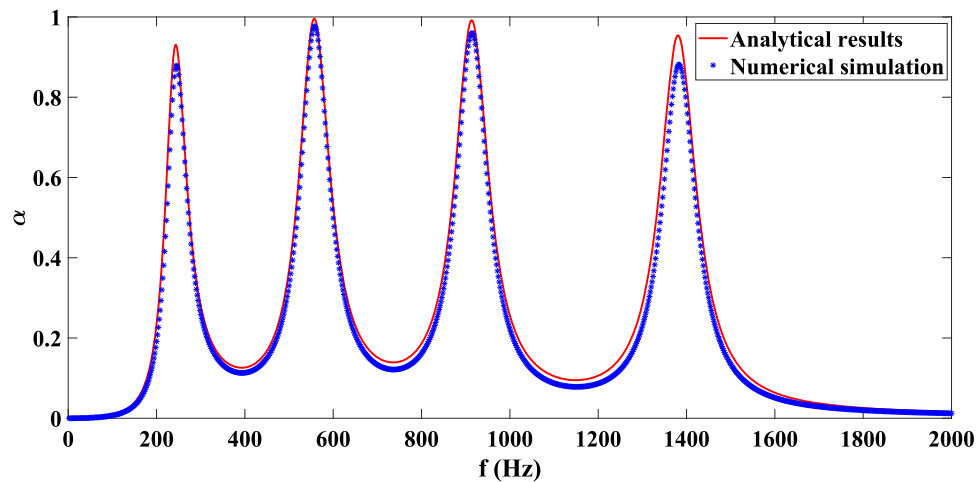


Fig. 9 Absorption coefficients obtained from analytical model and FEM simulation for series arrangement of dissimilar HRs in the descending order of neck diameter

2 mm, neck length $l_n = 8$ mm, cavity length $l_c = 12$ mm and cavity diameter $d_c = 16$ mm. Radius of the panel $r_d = 25$ mm. As expected the parallel assembly of similar HRs exhibited single absorption peak corresponding to the resonance condition. Comparison of analytically predicted absorption coefficient and FEM simulation results of this particular model is illustrated in Fig. 10. It is noticed that the analytical results are in good agreement with the FEM simulation results. In Fig. 11, the corresponding pressure plot of this model is depicted. The pressure plot shows significant sound power dissipation at resonance condition.

freq(315)=630 Hz Volume: Absolute pressure (Pa)

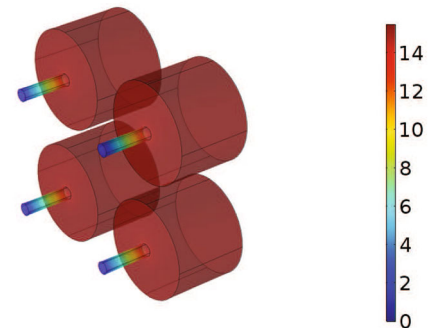


Fig. 11 Pressure plot of parallel configuration of similar HRs at resonance condition

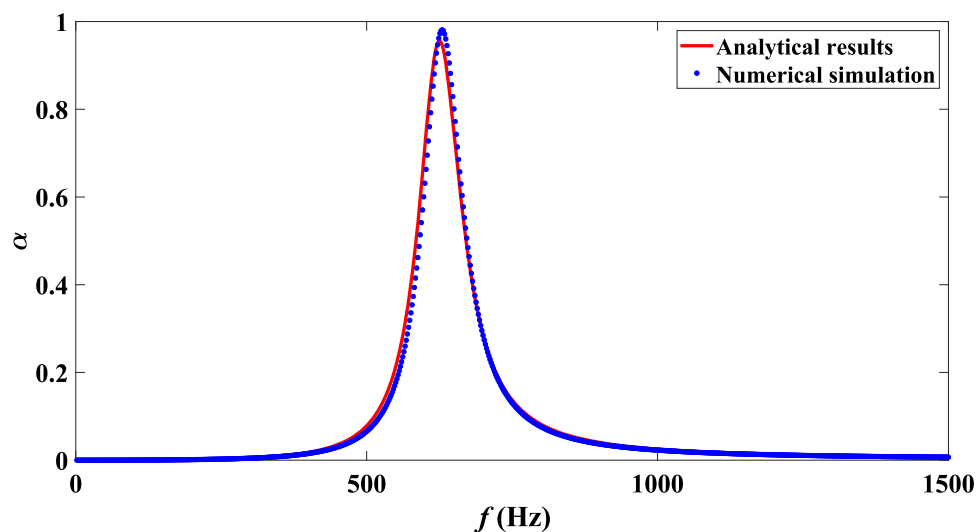


Fig. 10 Absorption coefficients obtained from analytical model and FEM simulation for parallel arrangement of similar HRs

Table 2 Parameters used for the parallel array of dissimilar HRs

Parameters	Value
Cavity diameter	16 mm
Cavity length	12 mm
Neck length	8 mm
Neck diameter	2 mm, 3 mm, 4 mm, 5 mm

3.3.2 Parallel Configuration of Dissimilar HRs

In the next case, parallel assembly of dissimilar HRs is considered. The dimensions used in the model are given in Table 2. The analytical results fit well with FEM simulation results as in Fig. 12. At the resonance frequency of individual resonators, significant sound absorption is occurring and that is evidently observed in the pressure plots (Fig. 13). Figure 12 also shows the anti-resonance effect, where sharp decline in sound absorption is observed. Anti-resonance is the condition which occurs when excited sound wave become out of phase with the resonator response. There should be a phase difference of 180 degree between the excitation and response at anti-resonance condition [23].

In Tables 3, 4, 5 and 6, the relative error between the FEM simulation results and analytical results of various configurations are given. From these, it can be observed that the relative error in resonance frequency is quite low in all cases. At the same time, the relative error in the maximum absorption coefficient is 9.75%, 9.1% and 9.37%, respectively, for series configuration of similar HRs, series configuration of dissimilar HRs and parallel configuration of dissimilar HRs.

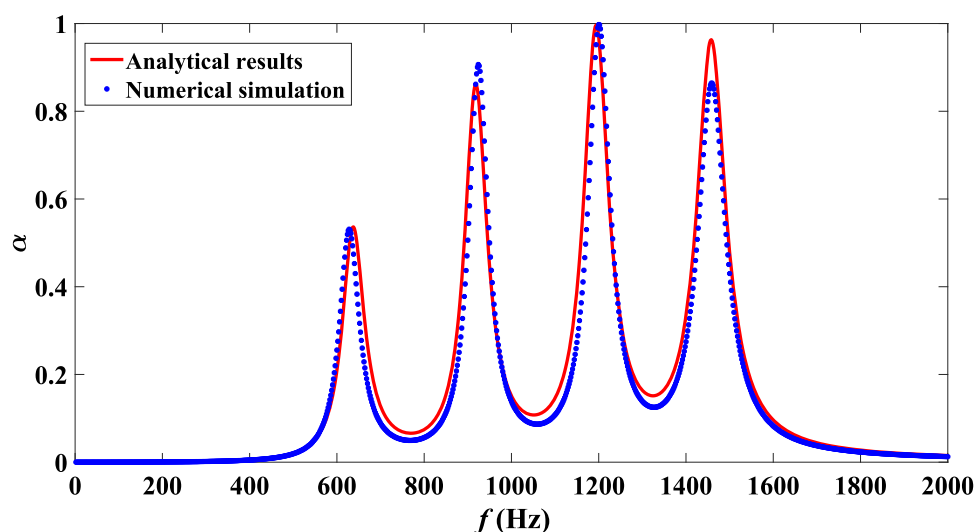
Table 3 Relative error between FEM result and analytical result for single HR and series configuration of similar HRs (frequency resolution 2 Hz)

	Models				
	HR	Series configuration of similar HRs			
		Peak1	Peak2	Peak3	Peak4
<i>Peak frequency</i>					
FEM	286	136	398	616	758
Analytical	286	134	396	614	756
Relative error(%)	0	1.5	0.5	0.32	0.26
<i>Maximum absorption coefficient</i>					
FEM	0.99	0.94	0.72	0.45	0.16
Analytical	0.97	0.92	0.68	0.41	0.15
Relative error(%)	2.06	2.17	5.88	9.75	6.67

Hence the FEM results fit well with the analytical results in all cases.

3.3.3 Tuning of HRs for Broad Band Sound Absorption

It is observed that the absorption band for the parallel arrangement of HRs (mentioned in Sect. 3.3.2), with randomly chosen dimensions, is extending from 628 to 1428 Hz. High anti-resonance drop between the resonance peaks is the major limitation noticed for the model. From simulations, it is noticed that the effect of anti-resonance can be minimized by properly tuning the dimensions of neck diameter. Hence for the better coupling of HRs and broad band sound absorption, certain combinations of neck diameters are simulated. The different configurations used are given in Table 7. All con-

**Fig. 12** Absorption coefficients obtained from analytical model and FEM simulation for parallel arrangement of dissimilar HRs

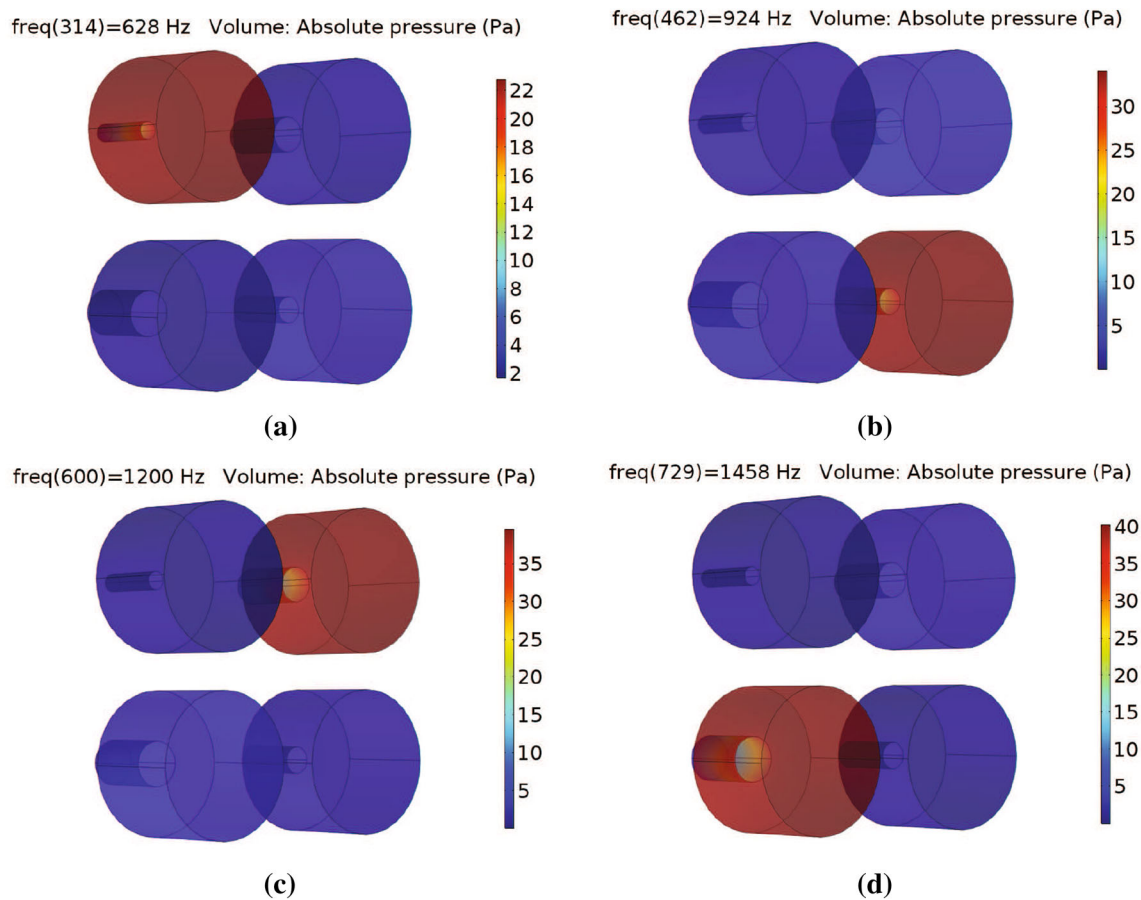


Fig. 13 Pressure plots of parallel configuration of dissimilar HRs at resonance frequencies **a** 628 Hz **b** 924 Hz **c** 1200 Hz and **d** 1458 Hz

Table 4 Relative error between FEM result and analytical result for series configuration of dissimilar HRs (ascending order of neck diameter) (frequency resolution 2 Hz)

	Model			
	Series configuration of dissimilar HRs (descending order of neck diameter)			
	Peak1	Peak2	Peak3	Peak4
<i>Peak frequency</i>				
FEM	166	636	1040	1530
Analytical	166	636	1042	1528
Relative error(%)	0	0	0.19	0.13
<i>Maximum absorption coefficient</i>				
FEM	0.997	0.72	0.14	0.0073
Analytical	0.999	0.66	0.13	0.0078
Relative error(%)	0.2	9.1	7.7	6.41

Table 5 Relative error between FEM result and analytical result for series configuration of dissimilar HRs (descending order of neck diameter) (frequency resolution 2 Hz)

	Model			
	Series configuration of dissimilar HRs (descending order of neck diameter)			
	Peak1	Peak2	Peak3	Peak4
<i>Peak frequency</i>				
FEM	246	558	914	1382
Analytical	244	558	914	1380
Relative error(%)	0.82	0	0	0.14
<i>Maximum absorption coefficient</i>				
FEM	0.88	0.98	0.96	0.88
Analytical	0.92	0.99	0.99	0.95
Relative error(%)	4.34	1.01	3.03	7.36

figurations are studied with neck length 8 mm, cavity length 12 mm and cavity diameter 16 mm. In the first configuration, (a), the neck diameters have a variation of 0.1 mm, which is about 1/1730 of minimum wavelength. In the remaining

configurations, (b to d), the neck diameters have variations of $\lambda_{\min}/865$, $\lambda_{\min}/576$, $\lambda_{\min}/432$ respectively. From the analysis, it is understood that when the difference between the diameter of necks increases the coupling between the res-

Table 6 Relative error between FEM result and analytical result for parallel configuration of similar & dissimilar HRs (frequency resolution 2 Hz)

	Models		Parallel configuration of dissimilar HRs			
	Parallel configuration of similar HRs		Peak1	Peak2	Peak3	Peak4
<i>Peak frequency</i>						
FEM	630		628	924	1200	1458
Analytical	626		638	918	1194	1458
Relative error(%)	0.63		1.56	0.65	0.5	0
<i>Maximum absorption coefficient</i>						
FEM	0.98		0.531	0.90	0.997	0.87
Analytical	0.96		0.536	0.86	0.996	0.96
Relative error(%)	2.08		0.93	4.65	0.1	9.37

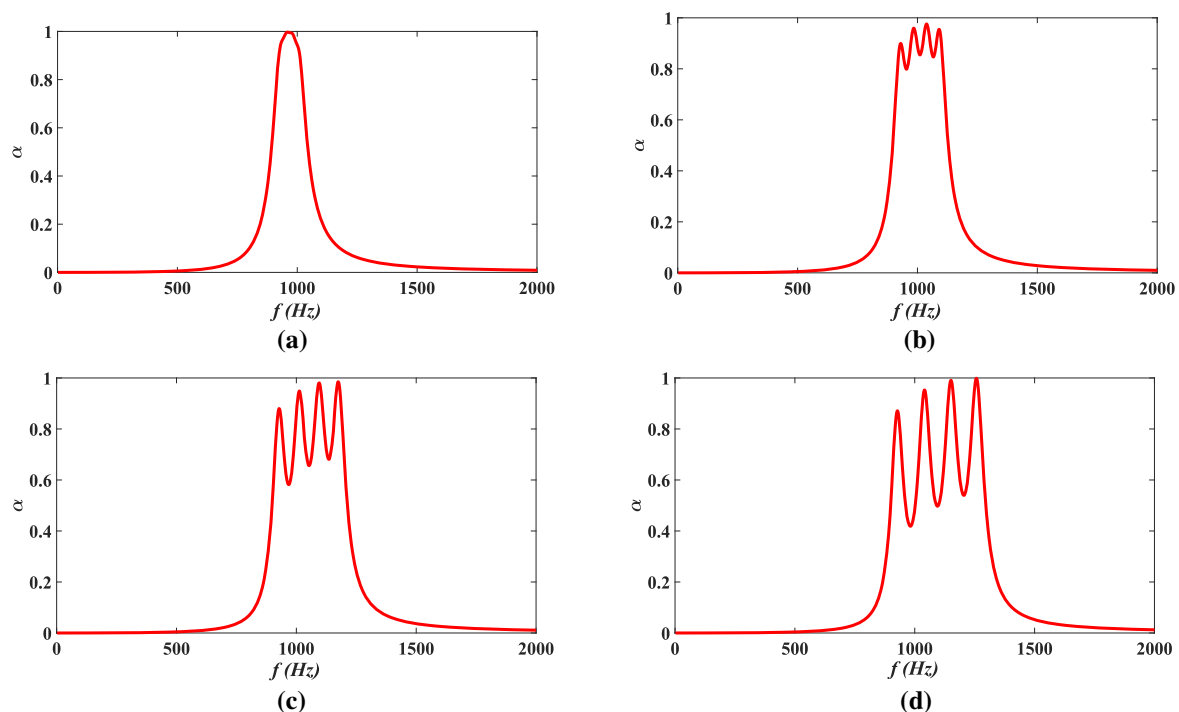
Table 7 Combination of neck diameters used for tuning

Configurations	Neck diameters
a	3 mm, 3.1 mm, 3.2 mm, 3.3 mm
b	3 mm, 3.2 mm, 3.4 mm, 3.6 mm
c	3 mm, 3.3 mm, 3.6 mm, 3.9 mm
d	3 mm, 3.4 mm, 3.8 mm, 4.2 mm

onators reduces (Fig. 14a–d), which in turn enhances the effect of anti-resonance. As depicted in Fig. 14a when the neck diameters have a variation of 0.1 mm ($\lambda_{\min}/1730$), all the four resonators coupled together and produce a single absorption peak, having slightly wide absorption bandwidth than the parallel configuration with same neck diameter (Fig. 10). Resonance peak corresponding to the individual resonators is found to emerge when the difference between the dimensions of neck diameter increases. The second configuration shows better absorption over a frequency band of 916–1106 Hz with more than 80% absorption. At the same time, the sound absorption of ‘c’ exceeds 50% from 900 to 1210 Hz, and the effective absorption bandwidth is extended to about two 1/3 octave bands. It is understood that by properly tuning the geometric parameters of the resonators, absorbers as per our requirement can be constructed.

3.4 Effect of Neck Position Offset on Absorption Coefficient and Resonance Frequency

In analytical formulations, the position of neck is not taken into account. But by numerical simulations, the effect of neck position on absorption characteristics can be investigated. The dimensions of HR chosen for this study are: neck diameter $d_n = 4$ mm, neck length $l_n = 8$ mm, cavity length $l_c = 12$ mm and cavity diameter $d_c = 50$ mm. The schematic diagram showing the position of neck in the HR is given in

**Fig. 14** Absorption characteristics of parallel array of HRs having variation in neck diameters tuned as **a** $\lambda_{\min}/1730$ **b** $\lambda_{\min}/865$ **c** $\lambda_{\min}/576$ **d** $\lambda_{\min}/432$

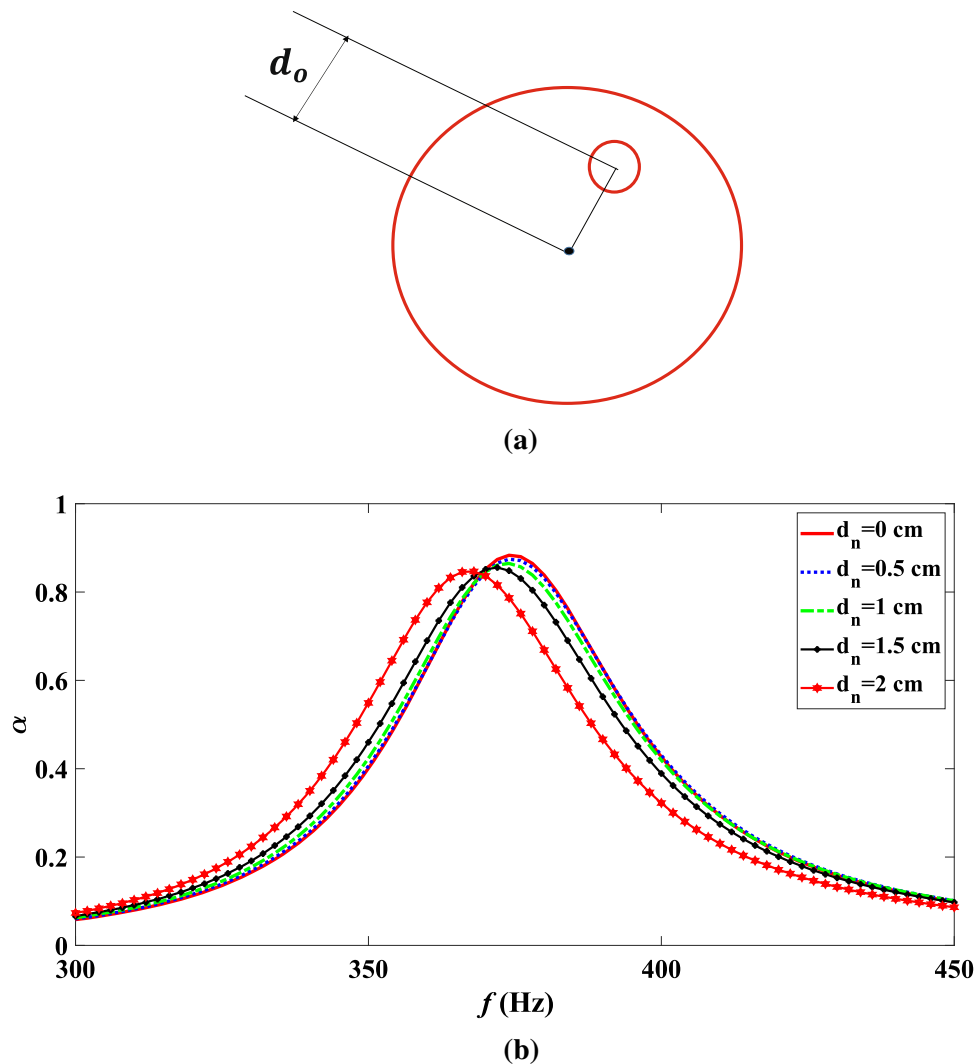


Fig. 15 **a** Schematic diagram of the HR with position of neck in the offset condition and **b** variation in sound absorption characteristics with position of neck

Fig. 15a, and the results corresponding to various neck positions are given in Fig. 15b. By increasing the offset distance of neck from the center (d_o), the maximum absorption coefficient reduces. The maximum absorption coefficient reduces by 4.2% when the neck position changes from center to an offset distance of 2 cm, whereas the resonance frequency varies by 2.14%. Hence with larger offset positions, absorption characteristics calculated using analytical formulations may lead to greater errors.

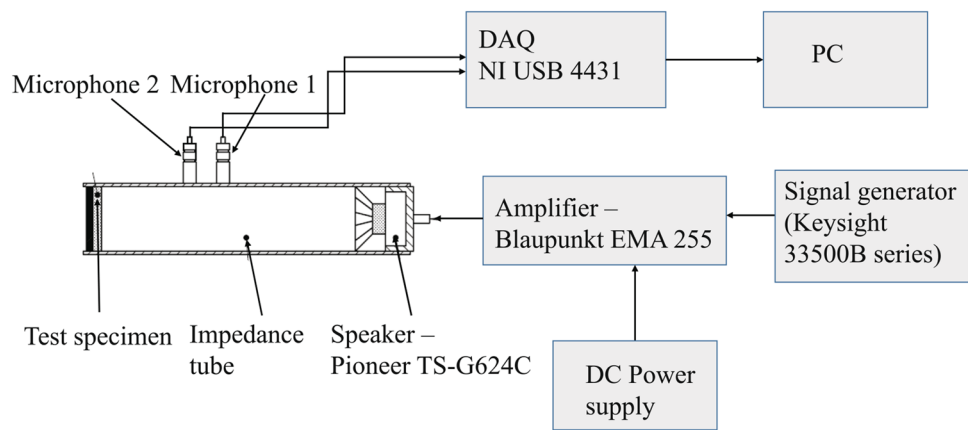
3.5 Experimental Result

In order to justify the numerical simulation results, absorption characteristics of one of the parallel assembly of HRs is experimentally determined. The dimensions of this model are given in Table 8. The experimental validation has been done

Table 8 Parameters of test sample

Parameters	Value
Cavity diameter	8 mm
Cavity length	3 mm
Neck length	17 mm
Neck diameter	2 mm, 3 mm, 4 mm, 5 mm

using two microphone impedance tube technique. Block diagram of the experiment setup is depicted in Fig. 16a. It constitutes an impedance tube, wave form generator, two microphones, speaker, amplifier, data acquisition system (DAQ) and a personal computer. The impedance tube is



(a)



(b)

Fig. 16 **a** Block diagram of experiment setup and **b** test sample

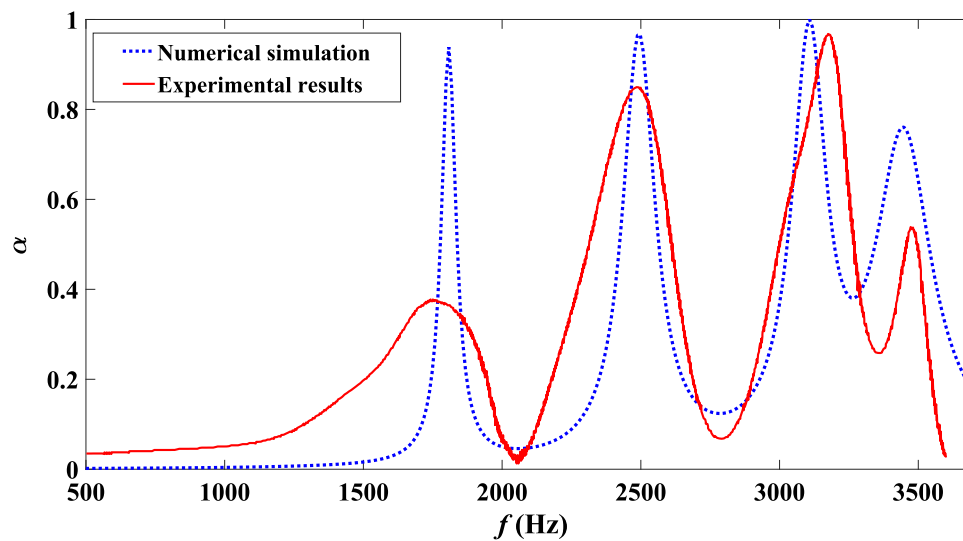


Fig. 17 Absorption coefficient against frequency for parallel array of HRs with varying neck diameters, from experiment and numerical simulation

designed and built as per ISO 10534-2 standard [51] in a cost effective manner.

The diameter of the tube is 25 mm and distance between the microphones is 4 cm. The test sample is mounted at the right end of the tube and a loudspeaker is placed at the left end to produce sound. The loud speaker was driven by a signal generator. From the signal generator, a sweeping pure tone of sinusoidal nature is applied as input in the frequency range of 300–3700 Hz. The signal is amplified using an amplifier. The pressure at two points is measured using two microphones of same type. MM310, 1/4 inch microphones are used and the sensitivity of microphones is 3.6 mV/Pa. The output signals are acquired and processed using NI-USB 4431 DAQ and LabVIEW software. The test samples for experiments are fabricated by fused deposition method (FDM) from poly lactic acid (PLA). Test sample is shown in Fig. 16b. The absorption coefficient obtained corresponding to different frequencies in the range is plotted in Fig. 17. Four peaks corresponding to the resonance frequencies of the four HRs are obtained in the experiment also. The experimental results show good agreement with the simulation results. In experiment and simulation, resonance frequencies are observed as almost same. The observed discrepancy in absorption coefficient between the simulated and numerical results may be due to the slight dimensional variations that might have occurred during fabrication of sample as well as due to the surface roughness of neck and cavity. In the numerical model, resonator was considered as rigid, and perfect reflection condition was provided on the internal walls of the resonator, which didn't consider damping caused by the internal surface roughness of the sample.

4 Conclusions

The normal incidence sound absorption characteristics of different configurations of HRs is studied analytically, numerically and experimentally. From these studies, following conclusions are derived.

- The relative error between the analytical and numerical results is very less.
- Parallel configuration of HRs resulted in the reduction of absorber thickness. Moreover, parallel arrangement of dissimilar HRs increased the absorption bandwidth.
- By proper tuning of geometric parameters and employing parallel configuration, the trade-off between wider band width and minimum space requirement can be overcome.
- The experimental observations of one of the parallel configuration is in-line with the numerical predictions.
- In most of the cases, analytical model matches with the FEM results. However, the analytical model has frequency constraints. Also the effect of position of HR neck

on absorption coefficient and resonance frequency is not taken into account in the analytical model, but it is correctly predicted by FEM simulations.

- The FEM also enables better visualization of acoustic absorption with frequency. Hence in practical implementations the software-based FEM calculations are more reliable to produce accurate predictions.

References

1. Jena, D., Dandsena, J., Jayakumari, V.: Demonstration of effective acoustic properties of different configurations of Helmholtz resonators. *Appl. Acoust.* **155**, 371–382 (2019)
2. Ingard, U.: On the theory and design of acoustic resonators. *J. Acoust. Soc. Am.* **25**(6), 1037–61 (1953)
3. Alster, M.: Improved calculation of resonant frequencies of Helmholtz resonators. *J. Sound Vib.* **24**(1), 63–85 (1972)
4. Munjal, M.L.: *Acoustics of Ducts and Mufflers with Application to Exhaust and Ventilation System Design*. Wiley (1987)
5. V  r, I.L., Beranek, L.L.: *Noise and vibration control engineering: principles and applications*. Wiley (2005)
6. Panton, R.L., Miller, J.M.: Resonant frequencies of cylindrical Helmholtz resonators. *J. Acoust. Soc. Am.* **57**(6), 1533–1535 (1975)
7. Tang, P., Sirignano, W.: Theory of a generalized Helmholtz resonator. *J. Sound Vib.* **26**(2), 247–262 (1973)
8. Selamet, A., Dicky, N.S., Novak, J.M.: Theoretical computational and experimental investigation of Helmholtz resonators with fixed volume : lumped versus distributed analysis. *J. Sound Vib.* **187**(2), 358–367 (1995)
9. Yoda, M., Sugiyama, S., Akishita, S., Konishi, S.: Tunable acoustic absorber with micro deep hole array. In: *Proceedings of the 1998 International Symposium on Micromechanics and Human Science*, pp. 177–182. IEEE (1998)
10. Prydz, R., Wirt, L., Kuntz, H., Pope, L.: Transmission loss of a multilayer panel with internal tuned Helmholtz resonators. *J. Acoust. Soc. Am.* **87**(4), 1597–1602 (1990)
11. Seo, S.H., Kim, Y.H.: Silencer design by using array resonators for low-frequency band noise reduction. *J. Acoust. Soc. Am.* **118**(4), 2332–2338 (2005)
12. Komkin, A.I., Mironov, M.A., Bykov, A.I.: Sound absorption by a Helmholtz resonator. *Acoust. Phys.* **63**(4), 385–392 (2017)
13. Kanev, N.G.: Maximum sound absorption by a Helmholtz resonator in a room at low frequencies. *Acoust. Phys.* **64**(6), 774–777 (2018)
14. Selamet, A., Lee, I.: Helmholtz resonator with extended neck. *J. Acoust. Soc. Am.* **113**(4), 1975–1985 (2003)
15. Tang, S.: On Helmholtz resonators with tapered necks. *J. Sound Vib.* **279**(3), 1085–1096 (2005)
16. Selamet, A., Xu, M.B., Lee, I.J., Huff, N.T.: Helmholtz resonator lined with absorbing material. *J. Acoust. Soc. Am.* **117**(2), 725–733 (2005)
17. Shi, X., Mak, C.M.: Helmholtz resonator with a spiral neck. *Appl. Acoust.* **99**, 68–71 (2015)
18. Cai, C., Mak, C.M., Shi, X.: An extended neck versus a spiral neck of the Helmholtz resonator. *Appl. Acoust.* **115**, 74–80 (2017)
19. Xu, M., Selamet, A., Kim, H.: Dual Helmholtz resonator. *Appl. Acoust.* **71**(9), 822–829 (2010)
20. Cai, C., Mak, C.M.: Hybrid noise control in a duct using a periodic dual Helmholtz resonator array. *Appl. Acoust.* **134**, 119–124 (2018)
21. Kim, S., Kim, Y.H., Jang, J.H.: A theoretical model to predict the low-frequency sound absorption of a Helmholtz resonator array. *J. Acoust. Soc. Am.* **119**(4), 1933–1936 (2006)

22. Tang, S., Ng, C., Lam, E.: Experimental investigation of the sound absorption performance of compartmented Helmholtz resonators. *Appl. Acoust.* **73**(9), 969–976 (2012)
23. Park, S.H.: Acoustic properties of micro-perforated panel absorbers backed by Helmholtz resonators for the improvement of low-frequency sound absorption. *J. Sound Vib.* **332**(20), 4895–4911 (2013)
24. Gai, X.L., Xing, T., Li, X.H., Zhang, B., Wang, W.J.: Sound absorption of microperforated panel mounted with Helmholtz resonators. *Appl. Acoust.* **114**, 260–265 (2016)
25. Mahesh, K., Mini, R.S.: Theoretical investigation on the acoustic performance of Helmholtz resonator integrated microperforated panel absorber. *Appl. Acoust.* **178**, 108012 (2021)
26. Fang, N., Xi, D., Xu, J., Ambati, M., Srituravanich, W., Sun, C., Zhang, X.: Ultrasonic metamaterials with negative modulus. *Nat. Mater.* **5**(6), 452–456 (2006)
27. Jiménez, N., Huang, W., Romero García, V., Pagneux, V., Groby, J.P.: Ultra-thin metamaterial for perfect and quasi-omnidirectional sound absorption. *Appl. Phys. Lett.* **109**(12), 121902 (2016)
28. Li, Y., Assouar, B.M.: Acoustic metasurface-based perfect absorber with deep subwavelength thickness. *Appl. Phys. Lett.* **108**(6), 063502 (2016)
29. Lan, J., Li, Y., Yu, H., Li, B., Liu, X.: Nonlinear effects in acoustic metamaterial based on a cylindrical pipe with ordered Helmholtz resonators. *Phys. Lett. A* **381**(13), 1111–1117 (2017)
30. Jiménez, N., Romero García, V., Pagneux, V., Groby, J.P.: Rainbow-trapping absorbers: broadband, perfect and asymmetric sound absorption by subwavelength panels for transmission problems. *Sci. Rep.* **7**(1), 1–12 (2017)
31. Yamamoto, T.: Acoustic metamaterial plate embedded with Helmholtz resonators for extraordinary sound transmission loss. *J. Appl. Phys.* **123**(21), 215110 (2018)
32. Hedayati, R., Lakshmanan, S.: Pneumatically-actuated acoustic metamaterials based on Helmholtz resonators. *Materials* **13**(6), 1456 (2020)
33. Long, H., Liu, C., Shao, C., Cheng, Y., Tao, J., Qiu, X., Liu, X.: Tunable and broadband asymmetric sound absorptions with coupling of acoustic bright and dark modes. *J. Sound Vib.* **479**, 115371 (2020)
34. Atak, O., Huybrechs, D., Pluymers, B., Desmet, W.: The design of Helmholtz resonator based acoustic lenses by using the symmetric multi-level wave based method and genetic algorithms. *J. Sound Vib.* **333**(15), 3367–3381 (2014)
35. Crow, B.C., Cullen, J.M., McKenzie, W.W., Koju, V., Robertson, W.M.: Experimental realization of extraordinary acoustic transmission using Helmholtz resonators. *AIP Adv.* **5**(2), 027114 (2015)
36. Yang, X., Yin, J., Yu, G., Peng, L., Wang, N.: Acoustic superlens using Helmholtz-resonator-based metamaterials. *Appl. Phys. Lett.* **107**(19), 193505 (2015)
37. Mahesh, K., Mini, R.S.: Helmholtz resonator based metamaterials for sound manipulation. *J. Phys: Conf. Ser.* **1355**, 012031 (2019)
38. Wang, T., Qin, Q., Zhu, X.: Reaction force and power flow analysis of an acoustic metamaterial beam with multi-band gaps. *Acoust. Aust.* **48**(1), 59–67 (2019)
39. Gu, J.C., Lin, W., Kan, C.X.: Sound source localization using piezoelectric acoustic metasurfaces. *Acoust. Aust.* **48**, 455–461 (2020)
40. Li, L., Liu, Y., Zhang, F., Sun, Z.: Several explanations on the theoretical formula of Helmholtz resonator. *Adv. Eng. Softw.* **114**, 361–371 (2017)
41. Chenzhi, C., Mak, C.M.: Noise attenuation capacity of a Helmholtz resonator. *Adv. Eng. Softw.* **116**, 60–66 (2018)
42. Johnson, D.L., Koplik, J., Dashen, R.: Theory of dynamic permeability and tortuosity in fluid-saturated porous media. *J. Fluid Mech.* **176**, 379–402 (1987)
43. Champoux, Y., Allard, J.F.: Dynamic tortuosity and bulk modulus in air-saturated porous media. *J. Appl. Phys.* **70**(4), 1975–1979 (1991)
44. Lauriks, W., Mees, P., Allard, J.F.: The acoustic transmission through layered systems. *J. Sound Vib.* **155**(1), 125–132 (1992)
45. Allard, J., Atalla, N.: *Propagation of Sound in Porous Media: Modelling Sound Absorbing Materials 2e*. Wiley (2009)
46. Verdière, K., Panneton, R., Elkoun, S., Dupont, T., Leclaire, P.: Transfer matrix method applied to the parallel assembly of sound absorbing materials. *J. Acoust. Soc. Am.* **134**(6), 4648–4658 (2013)
47. Verdière, K., Panneton, R., Elkoun, S., Dupont, T., Leclaire, P.: Comparison between parallel transfer matrix method and admittance sum method. *J. Acoust. Soc. Am.* **136**(2), 90–95 (2014)
48. Liu, S., Chen, W., Zhang, Y.: Design optimization of porous fibrous material for maximizing absorption of sounds under set frequency bands. *Appl. Acoust.* **76**, 319–328 (2014)
49. Abbad, A., Atalla, N., Ouisse, M., Doutres, O.: Numerical and experimental investigations on the acoustic performances of membraned Helmholtz resonators embedded in a porous matrix. *J. Sound Vib.* **459**, 114873 (2019)
50. Kaltenbacher, M.: *Computational Acoustics*. Springer (2018)
51. ISO 10534-2:1998(E): Determination of sound absorption coefficient and impedance in impedance tubes (1998)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.